

**LES RITO**

**MITSOUKA**

# **How data-driven recommendations can improve match quality while protecting user trust ?**

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# Who should we recommend?

Large catalogue of profiles



Shortlist shown to users

Random recommendations: 16.5% match rate

Model-based recommendations: 29.8% match rate in the top decile

Three candidate engines:

Lasso Logistic  
Regression

XGBoost

TabPFN

# What the model predicts ?

A match requires two independent decisions :

She says yes

AND

He says yes



Probability 1



Probability 2



Pair recommendation score

- The model predicts each person's decision separately
- The two probabilities are multiplied to obtain the pair score.
- This reflects the real product outcome: both users must be interested

# The data

Before

## Study from Columbia University

Source : Fisman, Iyengar, Kamenica & Simonson (2006), QJE — Speed Dating Experiment

- 8,378 rows
  - 4,184 pairs
  - 551 participants
  - 21 waves
  - 16.5% match
- 195 features

After

- Each date appears twice, once from each side → one row per pair, her\_\* / his\_\*
- 108 of the 195 columns are filled in during or after the date → leakage, removed
- Income is missing for 62% of Latino and 60% of Asian participants against 31% of Black ones

Target feature is

match

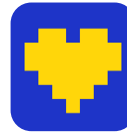
Protected attribute is

ethnicity

# Business model



Subscribes



Better recommendations



Retention  
+ recurring revenue

- Monthly subscription unlocks algorithmic recommendations instead of random matching
- The advantage should remain calibrated: enough to create value without making the user experience feel “solved” too quickly

- Recurring subscription revenue from premium users
  - Higher retention and lower churn through a more satisfying matching experience
- ➔ À remplacer par les vrais chiffres après :
- Estimate the inference cost of running the model per user and per month.
  - Benchmark against the share of users already paying for premium features on platforms such as Tinder or Hinge.
  - Set the subscription price above the variable recommendation cost while remaining competitive with existing premium plans.
  - Monitor the trade-off between recommendation performance, retention and profitability



# Statistical performance

| Engine                    | PR-AUC       | Top-decile match rate | Lift  |
|---------------------------|--------------|-----------------------|-------|
| Lasso logistic regression | 0.228        | 29.8%                 | 1.81x |
| XGBoost                   | 0.218        | 28.6%                 | 1.74x |
| TabPFN                    | <b>0.234</b> | 28.9%                 | 1.75x |

Peut-être plutôt mettre le bar chart avec le threshold (ce que Justin a fait sur le streamlit)

# Business value of a shortlist

At a 10% shortlist, the engine creates approximately 133 additional matches per 1,000 recommendations

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| Share of catalogue shown | Matches per 1,000 shown | Extra matches vs random |
|--------------------------|-------------------------|-------------------------|
| Top 5%                   | 314                     | +149                    |
| Top 10%                  | 298                     | +133                    |
| Top 20%                  | 235                     | +70                     |

# Performance

**16.5%**

Baseline Match  
Rate

**29.4%**

Top 10% Match Rate

**1.8x**

Conversion Lift

# Interpretability

| Feature                    | Coefficient | Std. Error | P-value  | Odds Ratio | Marginal Effect (%) | Scale / Step |
|----------------------------|-------------|------------|----------|------------|---------------------|--------------|
| female                     | -0.3142     | 0.0365     | < 0.0001 | 0.7304     | -13.74%             | 0 to 1       |
| self_race=Black            | 0.1124      | 0.0259     | < 0.0001 | 1.1190     | +11.40%             | 0 to 1       |
| self_goal=rare             | -0.0497     | 0.0260     | 0.0563   | 0.9515     | -10.68%             | 0 to 1       |
| other_field=social work    | -0.1145     | 0.0281     | < 0.0001 | 0.8918     | -10.41%             | 0 to 1       |
| self_field=social science  | 0.1023      | 0.0297     | 0.0006   | 1.1077     | +8.17%              | 0 to 1       |
| self_goal=relationship     | 0.0668      | 0.0254     | 0.0086   | 1.0691     | +7.95%              | 0 to 1       |
| other_field=education      | -0.0956     | 0.0297     | 0.0013   | 0.9088     | -7.72%              | 0 to 1       |
| self_field=education       | 0.0832      | 0.0299     | 0.0055   | 1.0867     | +6.97%              | 0 to 1       |
| other_goal=rare            | 0.0293      | 0.0512     | 0.5670   | 1.0297     | +6.69%              | 0 to 1       |
| samefield                  | 0.0914      | 0.0260     | 0.0004   | 1.0957     | +6.32%              | 0 to 1       |
| self_race=White            | -0.1387     | 0.0309     | < 0.0001 | 0.8705     | -6.12%              | 0 to 1       |
| other_field=social science | -0.0745     | 0.0293     | 0.0111   | 0.9282     | -5.77%              | 0 to 1       |
| other_race=rare            | 0.0219      | 0.0518     | 0.6727   | 1.0221     | +5.59%              | 0 to 1       |
| other_field=business       | 0.1031      | 0.0349     | 0.0031   | 1.1086     | +5.38%              | 0 to 1       |
| other_race=Asian           | -0.1018     | 0.0398     | 0.0106   | 0.9032     | -5.19%              | 0 to 1       |

# Stability

| Engine              | Model instability |
|---------------------|-------------------|
| Logistic regression | 0.56              |
| XGBoost             | 0.30              |

- Only around half of the top-decile shortlist is retained after retraining.
- Logistic regression: 48% retained
- XGBoost: 44% retained

Accuracy can remain stable while the exact recommendations change.

- Leave-one-wave-out PR-AUC range: approximately 0.10 to 0.46
- Wave size explains much of the variation.
- Anchoring the model to the previous version improve PR-AUC from 0.287 to 0.312.

→ Retrain with a stability constraint rather than replacing model completely at every update.

# Fairness

## Descriptive preference vs algorithmic impact

Observed user preferences  $\neq$  algorithmic fairness

- Asian participants appear in 5.8% of shown pairs versus 11.3% for others.
- Among actual matches, 10.5% of Asian pairs are surfaced versus 20.1% for others.
- Four out of six tested gaps remain after conditioning on relevant variables.



The question is not whether users have preferences. The question is whether the engine amplifies them.

# Measuring amplification

Amplification = share of same-background pairs shown ÷ share of same-background pairs that actually matched

- Same-background share among actual matches: 41%
- Same-background share among recommendations: 48%
- Amplification ratio: 1.16
- 95% confidence interval: [1.03, 1.30]

**The engine did not create the preference, but it amplified it.**

| Engine              | Amplification |
|---------------------|---------------|
| XGBoost             | 1.01          |
| Logistic regression | 1.16          |
| TabPFN              | 1.27          |

| Variant                             | Amplification | Top-decile match rate |
|-------------------------------------|---------------|-----------------------|
| Full model                          | 1.16          | 29.8%                 |
| Without ethnicity                   | 1.35          | 25.1%                 |
| Ethnicity neutralised at prediction | 1.57          | 24.8%                 |
| Full unawareness                    | 1.07          | 24.8%                 |

Removing ethnicity alone can make the outcome less fair, because proxy variables remain.

➔ A less amplifying recommendation policy costs approximately 50 matches per 1,000 recommendations.



# Recommendation to the client

1. Deploy the two-stage lasso logistic regression as the transparent baseline.
2. Anchor retraining to the previous model to reduce model drift.
3. Use XGBoost as a serious alternative if fairness is prioritised above interpretability.
4. Do not remove ethnicity blindly; monitor proxy effects instead.
5. Let the client choose the shortlist length based on the business value of a match.
6. Collect user interaction data to improve cold-start recommendations.
7. Re-run fairness tests as the user base grows.

**Pas convaincu de cette slide de claude**

# How the client can test the model

